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NAAC Accredited Autonomous Institution
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EEC1363-BATTERY MANAGEMENT SYSTEM

Smart Vehicle Event Data Recorder (SVEDR) using Internet of Things (IOT)

A PROJECT REPORT

Submitted by

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in partial fulfilment for the award of the degree

of

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in

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M.KUMARASAMY COLLEGE OF ENGINEERING, KARUR

(Autonomous Institution affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this project report **"THE SMART VEHICLE EVENT DATA RECORDER (SVEDR) USING THE INTERNET OF THINGS (IOT)"** is the bonafide work of " **POOBALAN.C (927623BEE069), SOUNDHAR.M (927623BEE096), VIKRAM.R (927623BEE119),SAKTHIVEL.A (927623BEE306)** " who carried out the project work during the academic year 2025-2026 under my supervision.

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This Project Work (EEC1363) report has been submitted for the (V) Semester Project viva voce Examination held on _____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

VISION AND MISSION OF THE INSTITUTION

VISION

To emerge as a leader among the top institutions in the field of technical education

MISSION

- Produce smart technocrats with empirical knowledge who can surmount the global challenges
- Create a diverse, fully-engaged learner-centric campus environment to provide quality education to the students
- Maintain mutually beneficial partnerships with our alumni, industry and professional associations.

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

VISION

To produce smart and dynamic professionals with profound theoretical and practical knowledge comparable with the best in the field.

MISSION

- Produce hi-tech professionals in the field of Electrical and Electronics Engineering by inculcating core knowledge.
- Produce highlycompetent professionals withthrust on research.
- Provide personalized training to the students for enriching theirskills.

PROGRAMMEEDUCATIONAL OBJECTIVES (PEOs)

PEO1: Graduates will have flourishing career in the core areas of Electrical Engineering and allied disciplines.

PEO2: Graduates will pursue higher studies and succeed in academic/research careers.

PEO3: Graduates will be a successful entrepreneur in creating jobs related to Electrical and Electronics Engineering /allied disciplines.

PEO4: Graduates will practice ethics and have habit of continuous learning for their success in the chosen career.

PROGRAMME OUTCOMES (POs)

After the successful completion of the B.E. Electrical and Electronics Engineering degree Program, the students will be able to:

- **PO1 Engineering knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals, and an engineering specialization to develop solutions to complex engineering problems.
- **PO2 Problem analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.
- **PO3 Design/development of solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.

- **PO4 Conduct investigations of complex problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.
- **PO5 Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.
- **PO6 The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.
- **PO7 Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.
- **PO8 Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
- **PO9 Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.
- **PO10 Project management and finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
- **PO11 Life-long learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

The following are the Program Specific Outcomes of Engineering Students:

- **PSO1:** Power Systems Engineering

  Apply comprehensive knowledge of electrical power systems to analyze, design and manage generation, transmission, and distribution networks, integrating emerging technologies and sustainable practices to contribute effectively to industry and societal needs.

- **PSO2:** Power Electronics and Drives

  Design and develop efficient, reliable power electronic systems and motor drives with an emphasis on sustainable energy use, and inclusive technology solutions that address societal and environmental challenges.

- **PSO3:** Electronics and Instrumentation Engineering

  Develop intelligent instrumentation and embedded systems for accurate measurements, monitoring and control in interdisciplinary domains addressing societal needs.

Abstract (Key words)	**POs Mapping**
Smart Vehicle Event Data Recorder, Internet of Things, Emergency Alert System, Driver Behaviour Analysis, Microcontroller-Based Safety System.	PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PSO1, PSO2, PSO3

SDG Goal		**Remarks**
SDG 9	Build resilient infrastructure, promote to the inclusive and sustainable industrialization and foster innovation	Enhances It enhances community safety and reduces disaster-related casualties, helping protectlives during emergencie.

ABSTRACT

The Smart Vehicle Event Data Recorder (SVEDR) using IoT technology is developed to strengthen road safety by continuously monitoring vehicle activity and driving behavior. It collects and analyzes real-time data, enabling early detection of unsafe conditions and accidents. When a critical event occurs, the system instantly sends alerts to support quick emergency response. The onboard processor interprets vehicle movements to accurately identify important driving actions. A manual control option ensures that drivers can override the automated system whenever necessary. Overall, SVEDR improves accident prevention, supports detailed post-incident analysis, and contributes to safer and more responsible transportation aligned with Sustainable Development .

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LIST OF ABBREVIATION

S.No	ABREVIATION	EXPANSION
1	EV	Electric vehicle
2	IOT	Internet of Things
3	RFID	Radio Frequency Identification
4	GPS	Global Positioning System
5	ICE	Internal combustion engine

LIST OF TABLE

S.No	ABREVIATION	EXPANSION
1	FR	Financial requirement

CHAPTER 1

INTRODUCTION

In recent years, the automotive industry has undergone a massive transformation with the introduction of electric vehicles (EVs), autonomous driving systems, and smart transportation technologies. These advancements have significantly improved driving comfort, reduced human errors, and paved the way for sustainable mobility. However, road accidents still remain a major global concern, causing injuries, loss of lives, and property damage. According to global traffic statistics, thousands of accidents occur daily due to factors such as reckless driving, mechanical failures, drowsiness, or intoxication. Although modern vehicles are equipped with advanced features like airbag systems, automatic braking, and lane detection sensors, there is still a pressing need for accurate event documentation and real-time emergency response.

Traditional event recorders or "black boxes" found in vehicles mainly focus on logging data before and after accidents. However, these devices lack real-time communication and are often inaccessible during emergencies. In most cases, accident data is either lost due to hardware failure or not communicated promptly to emergency services, resulting in delayed rescue operations. This situation underscores the necessity of a Smart Vehicle Event Data Recorder (SVEDR) — a system capable of automatically detecting abnormal events, recording essential vehicle parameters, and transmitting alerts instantly to concerned authorities or caregivers.

The Smart Vehicle Event Data Recorder (SVEDR) is an IoT-based innovation that aims to bridge the gap between accident detection, real-time alerting, and post-event data analysis. It combines multiple embedded sensors and communication modules to create an intelligent monitoring environment within

the vehicle. The system uses sensors such as alcohol, fire, temperature, and vibration sensors to continuously monitor the vehicle's condition and driver's state. For example, the alcohol sensor detects intoxicated driving behavior, while the fire sensor identifies ignition or overheating conditions that could lead to vehicle fires. A vibration or impact sensor plays a critical role in detecting collisions or accidents. These sensors collectively feed their data to a central microcontroller, such as Arduino, which acts as the processing unit of the system.

When any abnormal condition or emergency is detected, the SVEDR processes the sensor data and immediately transmits relevant information through an IoT communication module like GSM, Wi-Fi, or MQTT protocol. The transmitted data may include details like location coordinates, sensor readings, and event type, enabling emergency services or registered contacts to take immediate action. This combination of sensing, processing, and cloud connectivity forms a complete real-time monitoring ecosystem that enhances road safety and responsiveness.

The integration of IoT brings significant advantages to the system. Unlike traditional data recorders, IoT-enabled devices are capable of remote access, continuous data logging, and live tracking. The data can be visualized on an online dashboard or mobile application, allowing authorities or vehicle owners to monitor vehicle health and status. Moreover, the incorporation of IoT ensures scalability — enabling the collection of valuable big data insights from multiple vehicles, which can be used by traffic management systems to improve road safety policies and accident prediction models in the future.

The primary goal of developing the Smart Vehicle Event Data Recorder (SVEDR) is to create an affordable, efficient, and intelligent safety solution suitable for both modern and conventional vehicles. It helps establish

accountability, improve emergency response time, and analyze the root causes of accidents more accurately. The proposed system not only aids in saving lives but also contributes to developing smarter and safer transportation infrastructure.

1.1 PROBLEM STATEMENT

It was observed that current electric vehicles lacked a reliable system to monitor and record critical road events such as sudden impacts, unsafe driving, or collisions. This limitation made it difficult to analyze incidents accurately and caused delays in emergency response.

1.2 OBJECTIVE

1. To design and develop an IoT-based Smart Vehicle Event Data Recorder (SVEDR) capable of monitoring critical vehicle and driver parameters.
2. To detect accidents, unsafe driving behavior, and hazardous events such as fire, alcohol influence, or sudden impacts using integrated sensors.
3. To provide real-time alerts and location tracking through IoT modules, enabling quick communication with emergency services or family members.
4. To facilitate driver monitoring and obstacle detection for enhanced on- road safety.
5. To support post-accident analysis by recording and transmitting event data for future safety improvements.
6. To deliver a cost-effective, scalable, and reliable solution that enhances vehicle intelligence and contributes to smarter transportation systems.

CHAPTER 2

LITERATSURVEY

- Farooqetal.(2024):

  Proposed a real-time analytics system for electric vehicle (EV) safety using edge computing and IoT. Sensor data is processed locally before transmission to the cloud, ensuring faster response times and enhanced vehicle safety.
- Liuetal.(2023):

  Developed a cloud-based data recorder that logs EV-specific parameters such as battery temperature, regenerative braking behavior, and motor power output. The system supports predictive maintenance and early hazard detection.
- Pateletal.(2023):

  Introduced an integrated vehicle dynamics and EV-data recording system that captures motor performance, charge–discharge cycles, and driver behavior, allowing for comprehensive long-term safety analysis.
- Leeetal.(2022):

  Designed smart safety devices employing weight-based detection systems and audible alarms to identify abnormal weight distribution or distress conditions during faults or emergencies.
- Mehtaetal.(2021):

  Developed an accident detection and reporting system using IMU sensors and IoT connectivity. The solution provides real-time emergency alerts, monitors EV-specific conditions, and supports rapid rescue operations along with detailed incident analytics.

CHAPTER 3

PROJECT METHODOLOGY

3.1 PROPOSED WORK

The proposed system integrates multiple sensors and IOT technology to create a reliable, real-time vehicle event data recorder. It uses an Arduino UNO as the central controller, connected to various sensors for accident detection, driver monitoring, and vehicle safety checks.

- An IMU sensor detects sudden movements or impacts.
- An alcohol sensor monitors driver sobriety.
- An ultrasonic sensor detects nearby obstacles and abnormal battery swelling.
- A GPS module tracks the real-time location of the vehicle.
- Temperature and fire sensors monitor the battery condition and detect fire hazards.
- An SOS button and buzzer provide manual and audible alerts during emergencies.
- An IOT module (Node MCU) ensures instant data transmission to emergency services or concerned authorities.

This combination ensures real-time accident detection, emergency communication, and post-incident data recording. The proposed SVEDR system overcomes the limitations of existing models by offering multi-sensor integration, immediate alerts, and intelligent data analysis at a low cost, thereby enhancing vehicle safety and supporting smarter transportation infrastructure.The proposed work focuses on designing and developing an intelligent Smart Vehicle Event Data Recorder (SVEDR) that integrates sensor fusion, AI-based vision analytics, and IoT connectivity to enhance vehicle safety, accident detection, and real-time monitoring.

The system aims to overcome limitations of conventional vehicle monitoring units by combining multi-sensor data acquisition with intelligent decision-making capabilities.The core idea of the proposed system is to continuously monitor the vehicle's dynamic parameters using an IMU sensor, GPS module, alcohol detector, ultrasonic sensor, temperature/fire sensors, and wheel encoder inputs. These signals provide real-time information on vehicle movement, environmental hazards, driver condition, and potential collision risks. A microcontroller (Arduino/ESP32) serves as the control unit to process and synchronize all sensor outputs.

In addition to sensors, the system incorporates an AI-enabled vision module capable of detecting obstacles, fire/smoke, lane deviation, and unsafe driver behavior using lightweight CNN models. Visual data is fused with sensor readings to improve accuracy and reduce false alarms.The proposed solution enhances road safety by providing early hazard detection, automated emergency reporting, and post-incident data analysis. The SVEDR is designed to be cost-effective, reliable, and suitable for integration into both two-wheelers and four-wheelers. By leveraging sensor fusion, AI, and IoT, this work aims to develop a smart, connected, and efficient monitoring system that aligns with modern intelligent transportation requirements.The core idea of the proposed system is to continuously monitor the vehicle's dynamic parameters using an IMU sensor, GPS module, alcohol detector, ultrasonic sensor, temperature/fire sensors, and wheel encoder inputs.

These signals provide real-time information on vehicle movement, environmental hazards, driver condition, and potential collision risks. A microcontroller (Arduino/ESP32) serves as the control unit to process and synchronize all sensor outputs. In addition to sensors, the system incorporates an AI-enabled vision module capable of detecting obstacles, fire/smoke, lane deviation, and unsafe driver behavior using lightweight CNN models.

3.2 BLOCK DIAGRAM

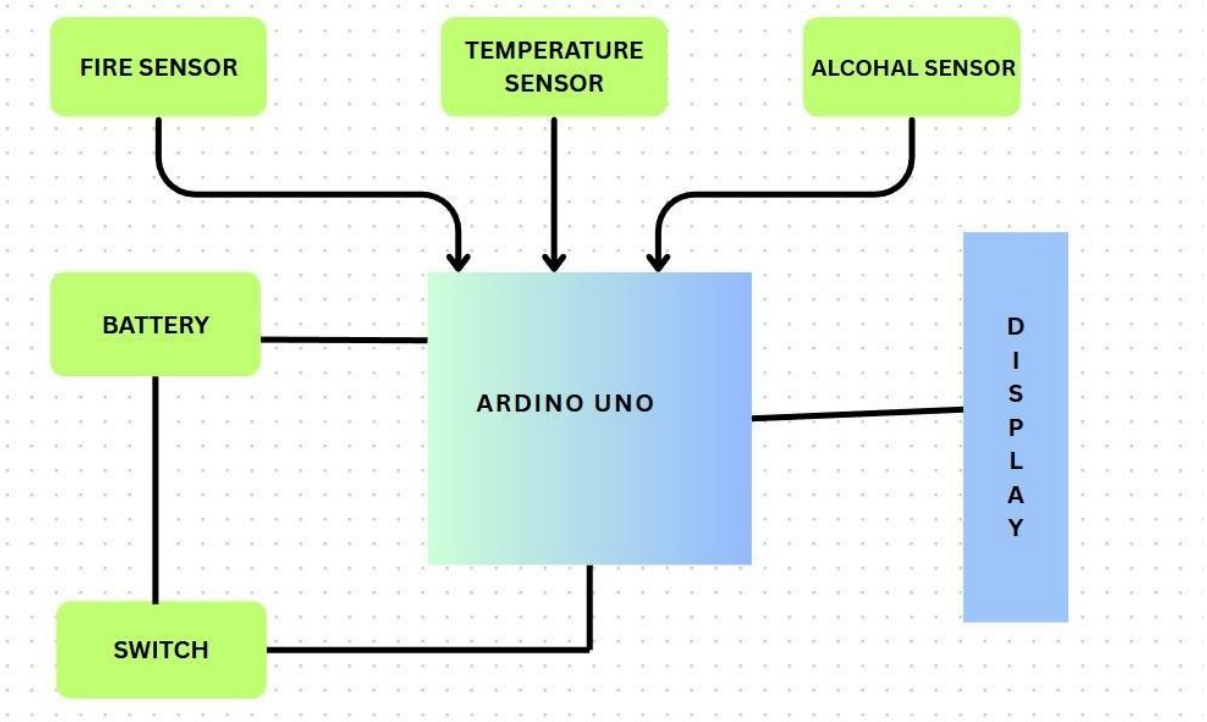

Figure 3.2 Block Diagram of Proposed System

3.2 BLOCK DIAGRAM DESCRIPTION

The block diagram of the Smart Vehicle Event Data Recorder (SVEDR) Zillustrates how multiple sensors and modules are integrated with the Arduino Uno to continuously monitor vehicle conditions. The system receives a stable 5V power supply, which energizes all components including the alcohol sensor, fire sensor, ultrasonic sensor, temperature sensor, IMU sensor, and GPS module. Each sensor plays a crucial role in detecting specific vehicle or environmental

parameters—such as alcohol level, fire hazards, distance from obstacles, vehicle tilt, acceleration, and real-time location. These sensors send their readings to the Arduino Uno, which acts as the central controller responsible for processing and analysing the collected data.

Based on the detected conditions, the Arduino triggers output actions through connected devices such as the buzzer, SOS button, and IoT module. In case of unsafe driving behaviour, fire detection, accident events, or manual SOS activation, the buzzer provides immediate audible warnings while the IoT module transmits emergency alerts and real-time GPS location data to cloud platforms or registered contacts. This enables rapid response and ensures the recorded data is securely stored for analysis. Overall, the block diagram represents a smart, IoT- enabled safety system designed to enhance vehicle monitoring, accident response, and driver safety.

Additionally, the integration of the IMU sensor and GPS module enables advanced event detection and accurate documentation of movement patterns. The IMU helps identify crashes, sudden tilts, or rollovers by measuring acceleration and angular velocity, while the GPS module logs the vehicle's location and speed continuously. This combination ensures that any critical event—not only accidents but also risky driving behaviours—is captured precisely and transmitted to the monitoring system. Such sensor fusion greatly improves the reliability and usefulness of the event data recorder.

Finally, the inclusion of an IoT communication module makes the system highly responsive and future-ready. The IoT unit uploads all processed data to cloud storage, enabling remote access for users, emergency services, or transportation authorities. This real-time connectivity allows for quick decision.

CHAPTER 4

PROTOTYPE OF THE MODEL

4.1 PROTOTYPE IMAGE

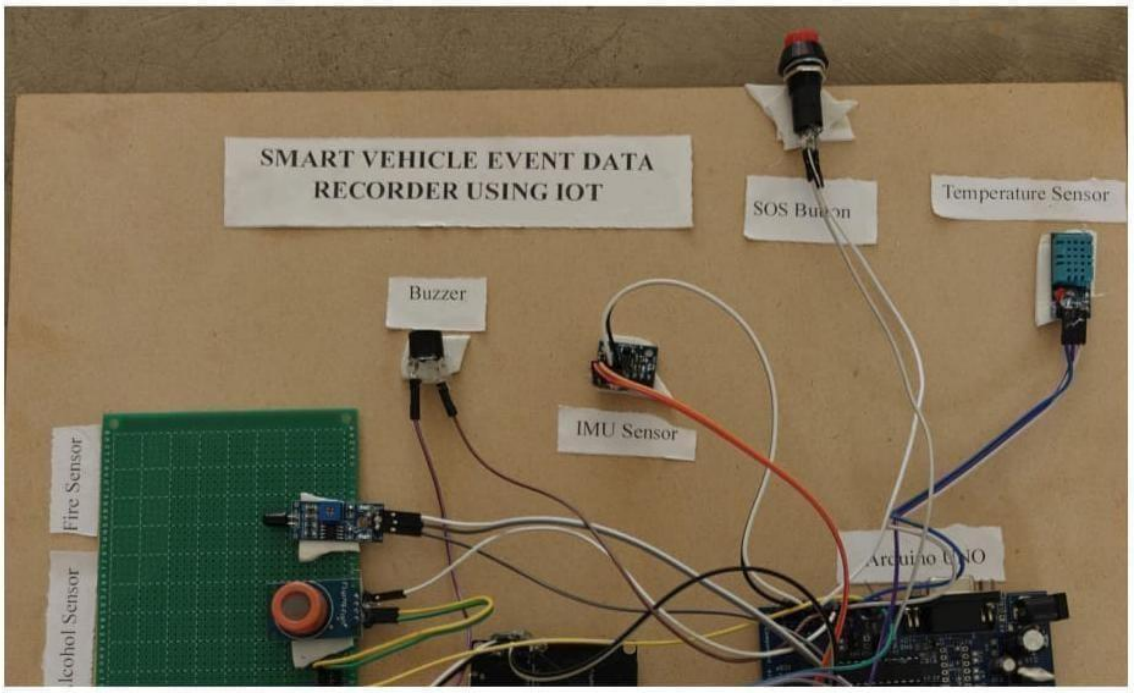

Figure: 3.1 HARDWARE KIT IMPLEMENTATION

4.2 PROTOTYPE EXPLANATION

The prototype shown in the image represents the practical implementation of the Smart Vehicle Event Data Recorder (SVEDR) using IoT. It is built using an Arduino Uno as the central controller, which collects data from multiple connected sensors. In the setup, the alcohol sensor, fire sensor, temperature sensor, ultrasonic sensor, and IMU sensor are arranged around the board, similar to how they would be distributed across real vehicle components. Each sensor

monitors a specific safety parameter — alcohol detecion for a driver sobriety, fire detection for onboard hazards, temperature monitoring for engine/ cabin conditions, and IMU readings for tilt, vibration, and impact detection. The wiring visible in the prototype demonstrates how these sensors feed continuous real-time signals into the Arduino for processing.

The prototype also includes an SOS button, clearly shown at the top right, which allows the driver or passenger to manually trigger an emergency alert. A buzzer, located near the IMU sensor, provides immediate audio notifications during unsafe conditions or accident detection. Once the Arduino interprets the sensor data, the connected IoT module (not visible but linked to the circuit) uploads the event data and GPS information to a cloud platform. This aligns with the abstract's objective of achieving real-time monitoring, emergency communication, and intelligent data recording.

This physical setup closely reflects how the system would perform inside a vehicle. The Arduino, acting as the system's CPU, integrates sensor data—similar to an onboard black box—and reproduces real-time accident and behavior detection scenarios. Although this prototype demonstrates the core sensing and alert mechanism, in a complete vehicle installation components like the GPS module, wheel encoder, indicator driver circuit, and ultrasonic positioning would be mounted at appropriate locations throughout the vehicle. These additions help determine turning angles, driving distance, crash impact, and vehicle orientation, as described in your abstract.

Overall, the prototype image demonstrates the essential functioning capability of the SVEDR: sensing, processing, alerting, and transmitting vehicle safety data. It proves the feasibility of the IoT-based design, showing how affordable incident analysis—strengthening road safety in alignment with Sustainable Development Goal 9 (IOT industrialization and foster innovation).

4.3 EXISTING MODEL

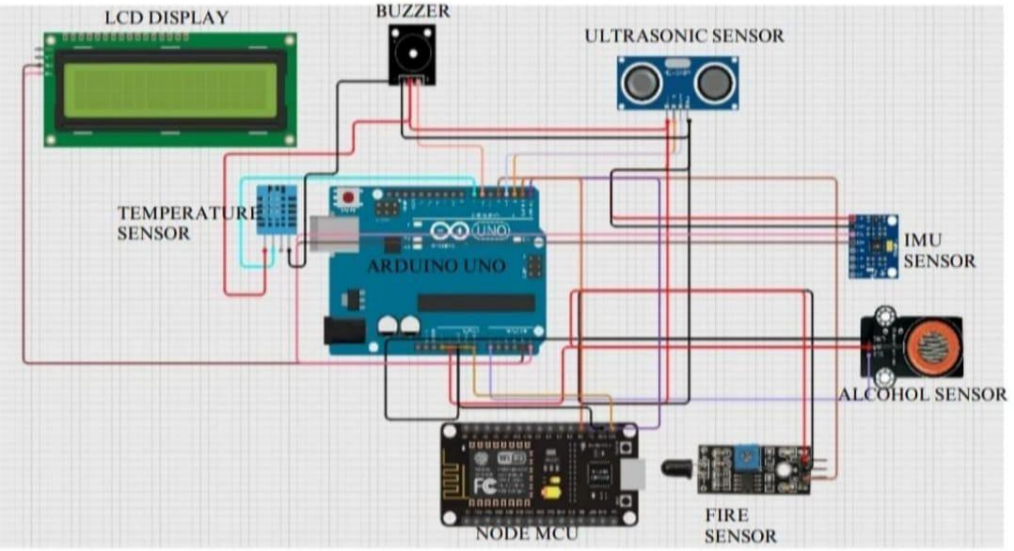

Figure: 4.3 SIMULINK OF EXISTING MODEL

CHAPTER 5

MODULE DESCRIPTION

5.1 DETAILED DESCRIPTION MODULE

The proposed Smart Vehicle Event Data Recorder (SVEDR) is an IOT- enabled safety system that integrates multiple modules to monitor vehicle and driver conditions in real time. At the core of the system, an Arduino UNO acts as the main controller, receiving inputs from various sensors and coordinating outputs. An IMU sensor detects sudden movements or impacts, helping in accident identification, while a GPS module tracks the exact location of the vehicle for emergency response. To ensure driver safety, an alcohol sensor monitors sobriety levels, and an ultrasonic sensor detects obstacles or signs of battery swelling. Additional safety modules include a temperature sensor to prevent overheating and a fire sensor to detect fire hazards. For communication, an IOT module (Node MCU/ESP8266) transmits sensor data and emergency alerts to cloud servers or designated authorities, ensuring quick response. The system also features an SOS button that allows passengers to trigger manual alerts, along with a buzzer and LCD display for audible and visual notifications. Together, these modules form a cost-effective, reliable, and intelligent solution that not only detects and reports accidents but also supports post-incident analysis, thereby contributing to safer and smarter transportation systems. device designed to enhance vehicle monitoring through the integration of multi-sensor data acquisition, AI-based vision analysis, and IoT communication. The system continuously collects real-time information using an IMU sensor for motion analysis, a GPS module for location tracking, an alcohol detector for driver sobriety monitoring, an ultrasonic sensor for obstacle detection, and temperature/fire sensors for identifying hazardous environmental conditions. A compact camera module, supported by an environmental of the

lightweight CNN algorithms, provides visual detection of lane deviation, obstacles, smoke, fire, and unsafe driver behavior, thereby improving accuracy through AI-driven interpretation. All sensor and visual data are processed by a microcontroller (Arduino/ESP32), which employs sensor fusion techniques to identify abnormal events such as collisions, sudden impact, excessive tilt, intoxicated driving, or fire outbreaks. In the event of a detected hazard, the system activates a buzzer or SOS mechanism and immediately communicates the event details—such as GPS location, time, and sensor values—to a cloud server via an IoT module, ensuring rapid notification to emergency contacts or authorities. The components are housed in a durable, vibration-resistant mechanical enclosure designed to withstand harsh vehicular conditions while maintaining operational stability. By combining robust sensing, intelligent processing, and real-time connectivity, the invention provides a compact, reliable, and cost-effective solution for improving road safety, accelerating emergency response, and enabling detailed post-incident analysis.

5.2 STRUCTURE OF MECHANICAL SYSTEMS

A plucking head mechanism with a servo-actuated gripper or cutter is located. Without damaging the plant or adjacent juvenile leaves, this mechanism may delicately grip and separate mature betel leaves. The robot has mounts for the camera, microprocessor, and battery pack and is frequently composed of The enclosure is typically constructed using high-strength ABS plastic or lightweight aluminum to provide impact resistance while keeping the overall system compact. Inside the housing, dedicated mounting slots are provided for the IMU sensor, GPS module, alcohol sensor, temperature/fire sensor, and the camera unit to ensure proper alignment and accurate data capture.

The camera is mounted with an adjustable bracket to maintain a stable field of view during vibration, movement, or sudden impacts.Anti-vibration pads or shock- absorbing mounts are placed beneath sensitive components like the IMU

camera module to reduce mechanical noise and improve detection accuracy. Ventilation slots are included to prevent overheating, especially when the IoT module or AI processing unit is active for extended periods. A separate compartment is provided for the power supply circuitry to minimize electromagnetic interference with sensor signals.The mechanical housing is designed to be weather-resistant, protecting the device from dust, moisture, and temperature fluctuations. Cable routing channels and sealed connectors ensure secure wiring and reduce wear due to continuous vehicle motion.

The compact form factor enables easy installation on various vehicle typeseither on the dashboard, chassis frame, or interior panels without affecting driver comfort or vehicle aesthetics.Overall, the mechanical structure ensures stability, protection, and reliable operation of the SVEDR, allowing the system to maintain data accuracy and performance under real-world vehicular conditions.

The mechanical structure of the Smart Vehicle Event Data Recorder (SVEDR) is designed to securely house all electronic components while ensuring durability, vibration resistance, and protection under harsh vehicle operating conditions. The mechanical system acts as the physical framework that supports, shields, and stabilizes sensors, the camera module, and the microcontroller.

5.3 AI DETECTION SYSTEM WITH VISION

A Raspberry Pi with a Pi camera, sometimes known as a ESP32- CAM module, is installed next to the plucking head. The betel leaves are photographed in real time by the camera. Based on color, size, and texture, a pre- trained AI model that can operate locally. It is trained to differentiate between mature and immature betel leaves. The AI system signals the plucking mechanism when it detects a mature leaf in the field of view. The vision unit consists of a compact camera module interfaced with the microcontroller or edge-AI hardware. Captured frames are processed using lightweight convolutional neural network (CNN) models optimized for embedded devices.

The system can detect visual cues such as lane deviation, obstacle proximity, fire, smoke, driver drowsiness, or unsafe driver behavior. The extracted features are fused with IMU, GPS, and environmental sensor data to produce a more reliable situational assessment. To achieve real-time detection under varied lighting and motion conditions, the module employs preprocessing techniques such as frame normalization, noise reduction, and edge extraction. Object detection and tracking pipelines (e.g., MobileNet-SSD, YOLO-Lite) help classify events while ensuring computational efficiency. When the AI vision module identifies a hazard or abnormal pattern, it triggers immediate alerts through the IoT communication layer, enabling cloud logging and emergency notification.

5.4 CONTROL LOGIC AND MICROCONTROLLERS

The central control unit is a Arduino Nano microcontroller. It Communicates with the AI camera module Regulates the servo motor according to the findings of detection Handles user inputs (such as manual override switches) controls power usage to prolong battery life.

5.5 POWER SUPPLY

A portable battery pack (such as a 12.4V rechargeable battery or a 7.4V Li- ion battery) powers the system. Proper voltage levels for the microprocessor and peripherals are guaranteed by a voltage regulator (such as the LM7805).

5.6 IOT FEATURES

The IoT feature of the Smart Vehicle Event Data Recorder (SVEDR) enables seamless connectivity, real-time communication, and remote data accessibility, making the system significantly more efficient than traditional standalone.

Through the integration of an IoT communication module (such as ESP32, GSM, Wi-Fi, or MQTT- based cloud platforms), the system continuously transmits sensor data— including IMU readings, GPS coordinates, alcohol levels, temperature, and impact signals—to a secure cloud server.

This connectivity allows critical events such as accidents, hazardous driving behavior, or abnormal sensor readings to be instantly communicated to emergency contacts, authorities, or monitoring systems. The IoT module ensures that alerts include precise location information, time stamps, and sensor metadata, enabling fast and informed decision- making during emergencies.

Additionally, the IoT feature supports remote logging, historical data retrieval, and analytics. Stakeholders can access vehicle status and event history through mobile or web dashboards, enhancing transparency and enabling post-incident investigation. The ability to remotely monitor the vehicle in real time also contributes to improved safety, reduced response time, and better compliance with road safety standard

5.7 BENEFITS

- Safety: Does not need climbing.
- Precision: Only mature leaves are targeted.
- Affordability: Makes use of inexpensive, readily available parts.
- Portability: Field deployable and lightweight.

A DC–DC buck converter (such as LM2596 or AMS1117) is used to step down the 12V input to the required operating levels while maintaining efficiency and minimizing heat generation. The converter ensures that voltage fluctuations from the vehicle battery—caused by engine ignition, acceleration, or load variations—do not affect the sensitive electronics. Additional filtering components, including capacitors and protection diodes, are used to suppress noise and prevent reverse polarity or voltage spikes.

The system also includes an optional rechargeable backup battery or supercapacitor, which provides temporary power in case of sudden vehicle shutdown or accident impact. This allows critical data, GPS coordinates, and event logs to be safely transmitted before the device powers off. The stable and isolated power supply

5.8 OPERATIONS

1. To arrive to the desired location, the user place the robot manually.
2. The AI camera finds mature betel leaves by scanning the area.
3. The controller activates the servo upon detecting a target leaf.
4. To harvest the leaf, the servo turns on the plucking mechanism.
5. The user continues the procedure after going to the next leaf.
6. Optional data is transferred or recorded for monitoring purposes.

The operation of the Smart Vehicle Event Data Recorder (SVEDR) is based on the coordinated functioning of sensors, the microcontroller, the AI vision unit, and the IoT communication module. Once the system is powered on, all sensors—including IMU, GPS, alcohol, ultrasonic, fire, and temperature—begin continuous monitoring of the vehicle's dynamic and environmental conditions. Each sensor generates real-time data, which is collected and processed by the microcontroller.

The AI vision module simultaneously captures video frames and analyzes them using lightweight machine learning models to detect visual cues such as obstacles, lane deviations, smoke, or driver fatigue. Sensor readings and visual data are fused to improve event detection accuracy. When the microcontroller identifies abnormal conditions—such as a sudden impact, intoxicated driving, fire, or high tilt angle—it immediately triggers local warnings through a buzzer and activates the SOS mechanism.

Parallel to the alert mechanism, the IoT communication module transmits key event data to a cloud server, including location, time, and detected hazards. This ensures that emergency contacts or authorities receive instant notifications for faster response. Normal operational data is also logged periodically for future analysis, enabling performance tracking and post-incident investigation.

CHAPTER 6

IMPLEMENTATION

6.1 EXPECTED TIME OF COMPLETION OF IDEA:

This timeline assumes access to basic resources, AI tools, and hardware components. For faster development, AI training and hardware assembly can proceed in parallel.

- Prototype Development (6 months): Finalize the prototype, enhancing it based on testing and feedback.
- Testing and Refinement (2 months): Conduct real-world tests to ensure effectiveness and reliability, refining design as needed.
- Production Preparation (2 months): Finalize manufacturing processes and source materials for production.
- Production and Launch (2 months): Begin mass production and launch the robot in the market.

6.2 FINANCIAL REQUIREMENTS:

Sl. No.	Description	Expected Expenditure
1	The cost of purchase of raw materials	1,000
2	The cost of contingencies	1,100
3	Product development cost	1,200
4	Other expenditure	1,200
Total		4,500

CHAPTER 7

CONCLUSION

The Smart Vehicle Event Data Recorder (SVEDR) integrated with AI-based vision and IoT technologies provides a comprehensive and intelligent solution for enhancing road safety and improving emergency response systems. By combining multi-sensor data acquisition, real-time video analysis, and cloud connectivity, the system is capable of accurately detecting accidents, hazardous conditions, and unsafe driver behavior. The fusion of IMU, GPS, alcohol, fire, ultrasonic, and temperature sensors ensures reliable monitoring of both vehicle dynamics and environmental factors. The AI vision module enhances event detection through visual interpretation, while the IoT communication unit enables instant transmission of critical data, such as crash alerts and location information, to emergency contacts or authorities. This significantly reduces response time and supports effective rescue operations. The system also facilitates post-incident analysis through continuous data logging. The overall design is compact, cost-effective, and easily adaptable to various vehicle types, making it suitable for widespread implementation. By addressing key limitations of traditional monitoring systems, the proposed SVEDR contributes to improved transportation safety and aligns with modern smart mobility and Sustainable Development Goals (SDGs), particularly SDG 3: Good Health and Well-Being. In conclusion, the project demonstrates how advanced sensing technologies, AI, and IoT can be integrated to create a reliable, intelligent, and user-friendly vehicle safety system capable of preventing accidents, minimizing damage, and ultimately saving lives.

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